The Mini Museum

TEACHER LESSON PLAN

Migration by Numbers

Grades 5–8 • Mathematics

GRADES 5–8 | 45 MINUTES | MATHEMATICS

LEARNING OBJECTIVE

What Students Will Understand

Students will use real Great Migration population data to practise percentage change, data interpretation, ratio and proportion, and graphical representation. They will understand that mathematics can reveal the scale of historical events — and that the numbers behind the Harlem Renaissance represent millions of individual human decisions.

STANDARDS ALIGNMENT

CCSS.Math 6.RP.A.3	Use ratio and rate reasoning to solve real-world problems; find percentage of a quantity
CCSS.Math 7.RP.A.3	Use proportional relationships to solve multi-step percent problems including percentage change
CCSS.Math 6.SP.B.4/5	Display and summarise numerical data; describe the nature of the attribute under investigation
CCSS.Math 8.SP.A.1	Construct and interpret scatter plots for bivariate measurement data; describe patterns

MATERIALS NEEDED

- Differentiated handouts — one per student
- Calculators (optional — Green level may benefit; Red level should work without)
- Graph paper or handout graph grids (included on handouts)
- Optional: project the Great Migration Map to give geographic context before the data work

HOOK: HOW BIG IS A MILLION? (5 MINUTES)

> *"Six million people moved north during the Great Migration. That's a number that's hard to visualise. Let's try: Holy Family has about 650 students. How many Holy Family schools would you need to fill to get 6 million students? Take 30 seconds and work it out."*

Answer: ~9,230 schools. Then: "Each one of those 6 million people made a decision — pack up, leave everything, travel north. Today we're going to look at what those decisions look like in numbers."

DIRECT INSTRUCTION: READING THE DATA (8 MINUTES)

Walk through the core dataset on the board — Chicago's Black population: 1910 (44,103), 1920 (109,458), 1930 (233,903), 1940 (277,731), 1950 (492,265), 1960 (812,637). Ask: before we calculate anything, what does this data tell us just by looking at it? What years show the fastest growth?

> *"Percentage change is the mathematician's way of making fair comparisons. A city growing by 10,000 people means something very different if it started with 50,000 versus 5 million. We calculate: (new − old) ÷ old × 100."*

ACTIVITY (25 MINUTES)

🟢 Green

Read a pre-drawn bar chart of Chicago's Black population growth. Identify the decade of fastest growth, calculate one percentage change with scaffolding, and answer interpretation questions in complete sentences.

🟡 Yellow

Calculate percentage change for all six decades. Create a line graph from the data table. Identify and explain the pattern. Compare Chicago to one other Northern city using a ratio.

🔴 Red

Full percentage change table for three cities (Chicago, New York, Detroit). Calculate rates of change. Analyse: which city's Black population grew fastest proportionally vs absolutely? Write a statistical summary paragraph and evaluate what the data does not show.

CLOSING (7 MINUTES)

> *"These numbers represent human beings. Every percentage point of growth is thousands of families who packed their belongings and moved north. Mathematics helps us understand the scale of history — but it can't tell us what it felt like. That's what the poets and painters were for."*

🏛 Mini Museum Connection

Option 1 — Visit the Exhibition: Bring your class to experience the phonograph, the Great Migration Map, the Jazz Lab, and more. Ask your librarian to arrange a visit. **Option 2 — Virtual Visit: minimuseumproject.com/exhibitions/harlem-renaissance**

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DATA TABLES — TEACHER REFERENCE

Black Population in Northern Cities (US Census Data)

City	1910	1920	1930	1940	1950	1960
Chicago, IL	44,103	109,458	233,903	277,731	492,265	812,637
New York, NY	91,709	152,467	327,706	458,444	747,608	1,087,931
Detroit, MI	5,741	40,838	120,066	149,119	300,506	482,223
Philadelphia, PA	84,459	134,229	219,599	250,880	376,041	529,240

WORKED EXAMPLE — PERCENTAGE CHANGE

Chicago 1910→1920: (109,458 − 44,103) ÷ 44,103 × 100 = 65,355 ÷ 44,103 × 100 = **148.2% increase**

Chicago 1950→1960: (812,637 − 492,265) ÷ 492,265 × 100 = 320,372 ÷ 492,265 × 100 = **65.1% increase**

Note: The 1910–1920 decade shows proportionally faster growth (148%) than the 1950–1960 decade (65%), even though the absolute increase was larger in 1950–1960. This is a key teaching moment for percentage change vs absolute change.

DISCUSSION QUESTIONS

- Detroit grew by 610% between 1910 and 1920. What was happening in Detroit's economy at that time that explains this? (Ford Motor Company, WWI armaments production)
- New York's Black population in 1960 was over 1 million. Harlem is one neighbourhood. What does this tell us about the limits of the "Harlem Renaissance" label?
- What does the data *not* show? (Living conditions, discrimination, wages, segregation in Northern cities)

CROSS-CURRICULAR CONNECTIONS

- **Social Studies:** Each spike in the data corresponds to a historical event — WWI, WWII, the Great Depression (note the slowdown in the 1930s), the post-war boom
- **Science:** Exponential vs linear growth; graph shapes and what they indicate about rate of change
- **ELA:** Writing a statistical summary paragraph requires the same claim-evidence-reasoning structure used in literary analysis

